

Hierarchical clustering

A general lattice theory perspective

Ali Siahkoohi¹, Joonas Ilmavirta², and Maarten V. de Hoop¹

¹Department of Computational Applied Mathematics and Operations Research, Rice University

²Department of Mathematics and Statistics, University of Jyväskylä

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“Flat” clustering

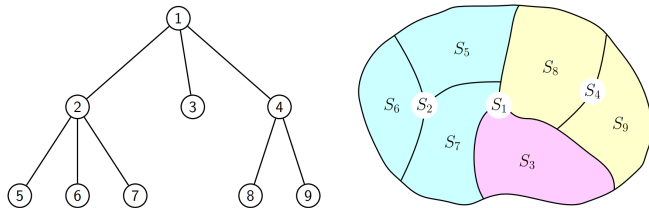
partitioning of data points into a fixed number of sets

The impossibility theorem of flat clustering

no clustering function simultaneously satisfies
scale invariance, richness, and consistency

Hierarchical clustering

family of partitionings of data points with varying degrees of “granularity”



Hierarchical clustering

relaxation of flat clustering: not restricted to a fixed number of clusters

no obstruction to existence (unlike flat clustering)

Gunnar Carlsson and Facundo Mémoli. "Characterization, Stability and Convergence of Hierarchical Clustering Methods". In: *Journal of Machine Learning Research* 11.47 (2010), pp. 1425–1470.

Benjamin Moseley and Joshua Wang. "Approximation Bounds for Hierarchical Clustering: Average Linkage, Bisecting K-means, and Local Search". In: *Advances in Neural Information Processing Systems*. Vol. 30. 2017.

Steinar Laenen, Bogdan Adrian Manghiuc, and He Sun. "Nearly-Optimal Hierarchical Clustering for Well-Clustered Graphs". In: *Proceedings of the 40th International Conference on Machine Learning*. Vol. 202. Proceedings of Machine Learning Research. July 2023, pp. 18207–18249.

Current hierarchical clustering algorithms

yield expected outcomes when applied to data with clear hierarchical structure

little is known theoretically when dealing with data with ambiguous discrete structure

impose unnecessary constraints on solution, e.g.

- ▶ tree (hyperbolic) metric embedding
- ▶ binary tree structure, e.g., single/average/complete linkage
- ▶ ultrametricity

Eli Chien, Puoya Tabaghi, and Olga Milenkovic. "HyperAid: Denoising in Hyperbolic Spaces for Tree-Fitting and Hierarchical Clustering". In: *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 2022, pp. 201–211.

Giovanni Chierchia and Benjamin Perret. "Ultrametric Fitting by Gradient Descent". In: *Advances in Neural Information Processing Systems*. Vol. 32. 2019.

Rishi Sonthalia and Anna Gilbert. "Tree! I am no Tree! I am a low dimensional hyperbolic embedding". In: *Advances in Neural Information Processing Systems*. Vol. 33. 2020, pp. 845–856.

A lattice-theoretic framework for hierarchical clustering

drawing inspiration from the work in cognitive psychology

stripping away unnecessary constraints and exploring the full solution space

Judith S Reitman and Henry H Rueter. "Organization Revealed by Recall Orders and Confirmed by Pauses". In: *Cognitive Psychology* 12.4 (1980), pp. 554–581.

Stephen C Hirtle. "Lattice-based similarity measures between ordered trees". In: *Journal of Mathematical Psychology* 25.3 (1982), pp. 206–225. DOI: 10.1016/0022-2496(82)90049-9.

Outline

hierarchical clustering in the context of lattice theory

- ▶ lattice theory
- ▶ the lattice of partition space
- ▶ hierarchical clustering as chain inference

information-theoretical energy landscape over the lattice of partitions

- ▶ minimum description length generative models
- ▶ optimal chain inference

computational heuristics

remaining challenges

Definition

a **lattice** L is a set X together with a relation $\leq$ that satisfies the following properties:

1. $\leq$ is a partial ordering, i.e., for all $x, y, z \in X$:
 - ▶ $x \leq x$
 - ▶ $x \leq y$ and $y \leq x \Rightarrow x = y$
 - ▶ $x \leq y$ and $y \leq z \Rightarrow x \leq z$
2. every pair of elements $x, y \in X$ has a unique least upper bound, i.e., join, denoted by $x \cup y$
3. every pair of elements $x, y \in X$ has a unique greatest lower bound, i.e., meet, denoted by $x \cap y$

Definition

a **partition** of set X is a collection $\mathcal{P}$ of non-empty subsets of X such that:

- ▶ every element of X belongs to exactly one subset in $\mathcal{P}$.
- ▶ the union of all subsets in $\mathcal{P}$ equals X , i.e., $\bigcup_{A \in \mathcal{P}} A = X$.
- ▶ any two distinct subsets in $\mathcal{P}$ are disjoint, i.e., $\forall A, B \in \mathcal{P}$, if $A \neq B$, then $A \cap B = \emptyset$.

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Definition

the **lattice of partitions** of set X , denoted $\mathcal{L}(\Pi(X), \leq)$, is the set of all partitions of X together with the relation $\leq$ such that for any two partitions $\mathcal{P}$ and $\mathcal{Q}$ of X , we say $\mathcal{P} \leq \mathcal{Q}$ if and only if for every $A \in \mathcal{P}$, there exists $B \in \mathcal{Q}$ such that $A \subseteq B$. This relation $\leq$ is called the **refinement ordering** on partitions.

The set $\Pi(X)$ together with the refinement ordering $\leq$ forms a lattice that satisfies the following properties:

1. $\leq$ is a partial ordering on $\Pi(X)$.
2. For any two partitions $\mathcal{P}$ and $\mathcal{Q}$ of X , their least upper bound (join) $\mathcal{P} \cup \mathcal{Q}$ is the coarsest partition that refines both $\mathcal{P}$ and $\mathcal{Q}$.
3. For any two partitions $\mathcal{P}$ and $\mathcal{Q}$ of X , their greatest lower bound (meet) $\mathcal{P} \cap \mathcal{Q}$ is the finest partition that is refined by both $\mathcal{P}$ and $\mathcal{Q}$.